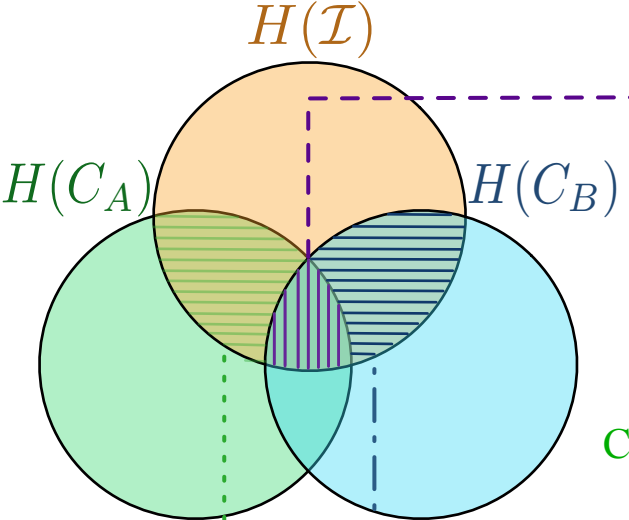

Shared information

$$S = I(\mathbf{x}_{\mathcal{I}}; \mathbf{x}_{\mathcal{C}_A}; \mathbf{x}_{\mathcal{C}_B})$$

Task-relevant information

$$R = \underbrace{C_A} + \underbrace{C_B} + \underbrace{S}$$

Caption A task-relevant information

$$C_A = I(\mathbf{x}_{\mathcal{I}}; \mathbf{x}_{\mathcal{C}_A} | \mathbf{x}_{\mathcal{C}_B})$$

Caption B task-relevant information

$$C_B = I(\mathbf{x}_{\mathcal{I}}; \mathbf{x}_{\mathcal{C}_B} | \mathbf{x}_{\mathcal{C}_A})$$